

Harry Nguyen

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SUMMARY

Full-stack engineer with production experience shipping TypeScript/React and C++ systems to 3B+ users in Chromium's continuous-deployment codebase. Primary engineer for backend, infra, and ML serving at TiMoto AI – 99.9% uptime, sub-50ms p99. Brings systems depth to product quality.

EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • GPA: 3.75/4.0

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

EXPERIENCE

Google | Software Engineer Intern – Chrome Browser

May 2025 – Aug 2025

- Architected **event-driven TypeScript/React system** with observer pattern decoupling UI state propagation – accelerated feature delivery **68%** across **25K+ lines** of Chromium at **95% test coverage**; changes reviewed by senior Chrome engineers and adopted into production branch
- Identified settings navigation as p99 bottleneck at 1,200ms; designed and implemented **lock-free concurrent trie search** eliminating mutex contention – **96% latency reduction**, zero production regressions; shipped to stable channel serving **3B+ active users**
- Designed **C++ IPC transport layer** with Protocol Buffers – selected Protocol Buffers over custom serialization for schema evolution; shipped to Chrome stable at **sub-50ms p99**, **10K+ req/sec**
- Collaborated with Chrome infrastructure team delivering design documents and code reviews at **95% test coverage**

TiMoto AI | Software Engineer

Sep 2025 – Present

- Led backend, cloud infrastructure, and ML serving for a 3-person engineering team – designed, deployed, and operated **distributed production systems** end-to-end, including full-stack product endpoints from Django REST API to TypeScript consumer clients
- Designed and maintained **gRPC inter-service layer**; resolved production deadlock under concurrent calls by tracing shared resource acquisition conflicts and redesigning call sequencing – **100% evaluation success rate** at **sub-50ms p99**
- Architected multi-AZ ECS Fargate with **Terraform IaC**, CloudWatch observability, and circuit breaker pattern – **99.9% uptime**, **44% cost reduction** (\$40–60/month); auto-rollback on health check failure
- Architected **vLLM inference engine with PagedAttention** for LLM serving – eliminated KV cache memory fragmentation; continuous batching achieving **zero OOM failures** at production traffic

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless BaaS on AWS – auto-scaling policy supporting **500+ concurrent users**, **90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+ on large datasets; redesigned with PostgreSQL indexing achieving **sub-100ms** for 10,000+ records

PROJECTS

Pulumi – Contributor | Go, TypeScript, IaC

github.com/pulumi/pulumi

- Submitted Go CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning – under active review by core maintainers of a **24K+ star** open-source project
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer – studied linearizability guarantees under concurrent operations

SKILLS

Languages: TypeScript, Python, Go, C++, JavaScript, Java, SQL, Bash, Rust

Frameworks & Databases: React, Django, FastAPI, Node.js, REST APIs, PostgreSQL, Redis, MongoDB

Distributed Systems: gRPC, Protocol Buffers, Circuit breakers, Exactly-once semantics, Kafka, Event-driven architecture

Cloud & Infrastructure: AWS (ECS Fargate, EKS, EC2, S3, RDS), Terraform, K8s, Docker, CI/CD, CloudWatch, Prometheus, Grafana

ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LoRA/QLoRA, PyTorch, LangChain, LLM-as-a-judge

AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor